

Africast-Time Series Analysis & Forecasting Using R

<https://workshop.f4sg.org/africast/>

9. Basic training and test accuracy



Outline

- 1 Evaluating forecast accuracy
- 2 Evaluating point forecast accuracy
- 3 Evaluating distributional forecast accuracy

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Evaluate forecast accuracy

- In order to evaluate the performance of a forecasting model, we compute its forecast accuracy.
- Forecast accuracy is compared by measuring errors based on the test set.
- Ideally it should allow comparing benefits from improved accuracy with the cost of obtaining the improvement.

Evaluate forecast accuracy- Business impact

- We should be choosing forecast models that lead to better business decisions
 - ▶ least staffing costs, least emission, highest service level, least stock-out, least inventory, fastest response, least change in planing, for example.
- However, this is not always easy to obtain, therefore we might simply use methods that provide the most accurate forecast.

In-sample (training) vs. out-of-sample (test)

- Fitting and its residual are not a reliable indication of forecast accuracy
- A model which fits the training data well will not necessarily forecast well
- A perfect fit can always be obtained by using a model with enough parameters
- Over-fitting a model to data is just as bad as failing to identify a systematic pattern in the data

Forecast accuracy evaluation using test sets

- We mimic the real life situation
- We pretend we don't know some part of data (new data)
- It must not be used for *any* aspect of model training
- Forecast accuracy is computed only based on the test set

Training and test sets



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Forecast errors

Forecast “error”: the difference between an observed value and its forecast.

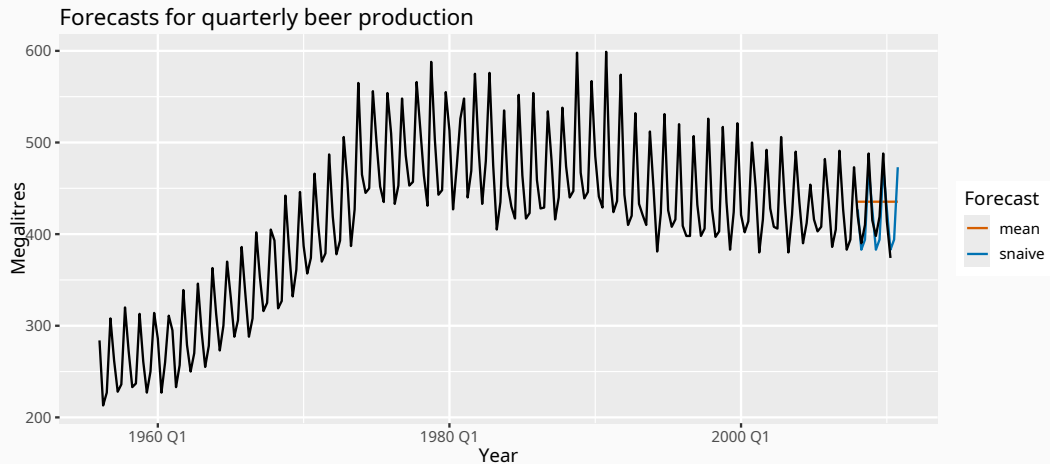
$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T},$$

where the training data is given by $\{y_1, \dots, y_T\}$

Measures of forecast accuracy

```
beer_fit <- aus_production |>
  filter(between(year(Quarter), 1992, 2007)) |>
  model(
    snaive = SNAIVE(Beer),
    mean = MEAN(Beer)
  )
beer_fit |>
  forecast(h = "3 years") |>
  autoplot(aus_production, level = NULL) +
  labs(title = "Forecasts for quarterly beer production",
       x = "Year", y = "Megalitres") +
  guides(colour = guide_legend(title = "Forecast"))
```

Measures of forecast accuracy



Measures of forecast accuracy

y_{T+h} = $(T + h)$ th observation, $h = 1, \dots, H$

$\hat{y}_{T+h|T}$ = its forecast based on data up to time T .

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

$$\text{MAE} = \text{mean}(|e_{T+h}|)$$

$$\text{MSE} = \text{mean}(e_{T+h}^2)$$

$$\text{RMSE} = \sqrt{\text{mean}(e_{T+h}^2)}$$

$$\text{MAPE} = 100\text{mean}(|e_{T+h}|/|y_{T+h}|)$$

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RMSE = $\sqrt{\text{mean}(e_{T+h}^2)}$

MAPE = $100\text{mean}(|e_{T+h}|/|y_{T+h}|)$

- MAE, MSE, RMSE are all scale dependent.
- MAPE is scale independent but is only sensible if $y_t \gg 0$ for all t , and y has a natural zero.

Measures of forecast accuracy

Mean Absolute Scaled Error

$$\text{MASE} = \text{mean}(|e_{T+h}|/Q)$$

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- For non-seasonal series, scale uses naïve forecasts:

$$Q = \frac{1}{T-1} \sum_{t=2}^T |y_t - y_{t-1}|$$

- For seasonal series, scale uses seasonal naïve forecasts:

$$Q = \frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|$$

where m is the seasonal frequency

Measures of forecast accuracy

Mean Absolute Scaled Error

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where m is the seasonal frequency

Proposed by Hyndman and Koehler (IJF, 2006).

Measures of forecast accuracy

Root Mean Squared Scaled Error

$$\text{RMSSE} = \sqrt{\text{mean}(e_{T+h}^2/Q)}$$

- For non-seasonal series, scale uses naïve forecasts:

$$Q = \frac{1}{T-1} \sum_{t=2}^T (y_t - y_{t-1})^2$$

- For seasonal series, scale uses seasonal naïve forecasts:

$$Q = \frac{1}{T-m} \sum_{t=m+1}^T (y_t - y_{t-m})^2$$

where m is the seasonal frequency

Proposed by Hyndman and Koehler (IJF, 2006).

Measures of forecast accuracy

```
beer_fc <- forecast(beer_fit, h = "3 years")  
accuracy(beer_fc, aus_production)
```

```
# A tibble: 2 x 10
```

	.model	.type	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	mean	Test	-13.8	38.4	34.8	-3.97	8.28	2.20	1.96	-0.0691
2	snaive	Test	5.2	14.3	13.4	1.15	3.17	0.847	0.729	0.132

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Prediction interval accuracy using winkler score

Winkler proposed a scoring method to enable comparisons between prediction intervals:

- it takes account of both coverage and width of the intervals.

Winkler score

$$W(l_t, u_t, y_t) = \begin{cases} u_t - l_t & \text{if } l_t < y_t < u_t \\ (u_t - l_t) + \frac{2}{\alpha}(l_t - y_t) & \text{if } y_t < l_t \\ (u_t - l_t) + \frac{2}{\alpha}(y_t - u_t) & \text{if } y_t > u_t \end{cases}$$

Prediction interval accuracy

```
# Compute interval accuracy  
beer_fc |> accuracy(aus_production,  
  measures = interval_accuracy_measures)
```

```
# A tibble: 2 x 5
```

	.model	.type	winkler	pinball	scaled_pinball
	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	mean	Test	174.	8.67	0.274
2	snaive	Test	86.3	3.10	0.0980

Quantile score

Quantile score

$$Q_{p,t} = \begin{cases} 2(1-p)(f_{p,t} - y_t), & \text{if } y_t < f_{p,t} \\ 2p(y_t - f_{p,t}), & \text{if } y_t \geq f_{p,t} \end{cases}$$

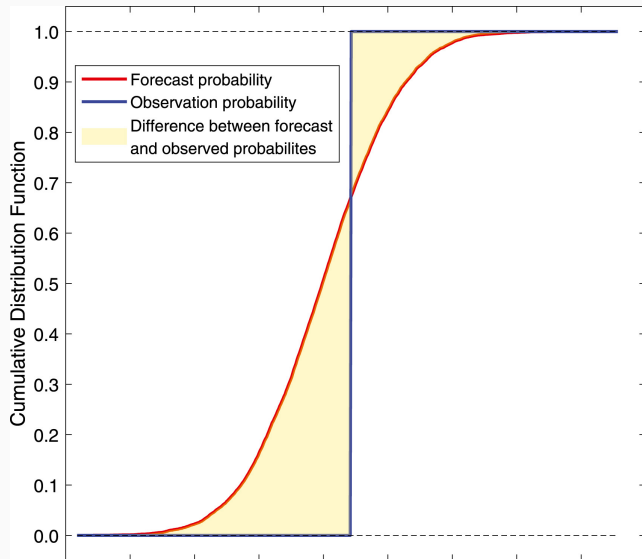
Continuous Ranked Probability Score (CRPS)

$$\text{CRPS} = \text{mean}(p_j),$$

where

$$p_j = \int_{-\infty}^{\infty} (G_j(x) - F_j(x))^2 dx,$$

Continuous Ranked Probability Score (CRPS)



Quantile score and CRPS

```
beer_fc |>  
  accuracy(aus_production, list(measures=distribution_accuracy_measures))
```

```
# A tibble: 2 x 4
```

	.model	.type	measures.percentile	measures.CRPS
	<chr>	<chr>	<dbl>	<dbl>
1	mean	Test	22.7	22.4
2	snaive	Test	8.80	8.71

Quantile score

```
beer_fc |>  
  accuracy(aus_production, list(qs=quantile_score), probs = .9)
```

```
# A tibble: 2 x 3  
  .model .type    qs  
  <chr>  <chr> <dbl>  
1 mean   Test   14.1  
2 snaive Test    4.60
```